

Mapping center pivot irrigation systems in global arid regions using instance segmentation and analyzing their spatial relationship with freshwater resources

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ABSTRACT

Ensuring freshwater availability and supply sufficient for socio-economic development is important for human health and productivity. Globally, increasing industrial agriculture is driving crop productivity, with increasing surface water and groundwater being withdrawn for irrigation. This study mapped center pivot irrigation systems (CPIS), which are indicators of intensified industrial agriculture and large-scale investment in agriculture, with satellite data using a convolutional neural network in global arid areas. A cascade instance segmentation network was adopted, and both the convolutional block attention module technique and PointRend technique were used to increase the performance of off-the-shelf algorithms. Geospatial analysis methods were used to derive the relationships between CPIS and freshwater resources. Overall there were about 10.26 million hectares (Mha) of irrigated areas equipped with CPIS in global arid areas in 2021, and there were 4.41 Mha of CPIS-equipped irrigated fields added in the past twenty years at a relative increase of 75% compared with those at the beginning of this century. Although the largest extent of increased CPIS-equipped irrigated fields was in North America, the other five continents had higher relative increases, with Africa having the highest relative growth of 253%. The analysis of spatial relationships between CPIS and freshwater resources indicates that the stress on freshwater resources from industrial agriculture is intensifying in arid areas globally. Overall, the world's arid regions with the highest water stress from CPIS are mostly located in western North America, northern Africa, and the Arabian Peninsula. The results facilitate a better understanding of the current status of freshwater stress and ecosystem sustainability in relation to the development of agricultural intensification in global arid areas. The outcomes support governments to judge better the environmental consequences of agricultural modernization and further help them make balanced agriculture and water management policies.

1. Introduction

Freshwater is one of the most precious and threatened resources for human beings and terrestrial organisms. With economic growth, urbanization, and agriculture expansion, freshwater demand continues to grow globally (UNESCO, 2021). However, the quality and quantity of freshwater resources are degrading globally, especially in developing countries (UNESCO, 2020). This presents a big challenge for ensuring enough freshwater availability and supply for sustainable development, which is important for human health and productivity. This calls for

global sustainable water management (Gleeson et al., 2020).

In arid areas (including all climatic hyper-arid, arid and semi-arid areas if not specified in the following), farmers generally need to use irrigation to increase crop productivity because of physical water scarcity and relatively low rainfall. Irrigated agriculture has a large benefit in that it is generally more productive than rain-fed agriculture. Water for irrigation is generally withdrawn from nearby surface water bodies (e.g., from rivers, lakes or reservoirs) in water-rich areas. However, in many arid areas around the world, such as in the western United States, South Africa and Arabian Peninsula, groundwater is the primary source

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of water supply for irrigation, especially during dry periods (Nhamo et al., 2020; Youssef et al., 2020). For example, southwestern North America is experiencing the most severe mega-drought in 1200 years (Liu et al., 2022; Williams et al., 2022). Groundwater is a valuable renewable resource. However, sustained excessive groundwater pumping for irrigation has caused groundwater depletion around the world where the rate of extraction exceeds the rate of recharge (de Graaf et al., 2019; Wada et al., 2012). Excessive groundwater pumping can cause detrimental effects, such as lowered water tables, reduced baseflow to streams and lakes, degradation of groundwater-dependent ecosystems, land subsidence, salt-water intrusion, and increased costs for the user (Barlow and Reichard, 2010; Chen et al., 2019c; Konikow, 2013).

In fact, uncontrolled freshwater withdrawal has caused many ecological, economic, and political problems. For example, the semi-arid High Plains region in the United States, which depends on the Ogallala Aquifer, is one of the most agriculturally productive regions in the world. However, parts of the Ogallala Aquifer, which is one of the world's largest aquifers, have been depleted after sixty years of intensive farming (Dennehy, 2000). For example, the Ogallala aquifer has a groundwater withdrawal of 10.1 mm/yr (Richey et al., 2015). In the Kingdom of Saudi Arabia, there has been an increasing number of sinkholes induced by groundwater level decline, which are associated with the growth of irrigated crops in the past two decades (Youssef et al., 2020). In Zambia, with the expansion of sugarcane fields from 2006, there is an obvious increase in irrigation water extraction from the Kafue River, which could have detrimental impacts on downstream users and ecosystems (German et al., 2020). In Central Asia, intense irrigation has caused the Aral Sea to lose almost 90% of its water surface and changed the regional climate (He et al., 2021). In California, USA, there has been a consistent significant deterioration of riparian woodland health associated with groundwater decline during the 2012–2019 California drought (Kibler et al., 2021). Growing water shortages also contribute to some rising regional tensions and conflicts worldwide over scarce water resources (Gettleman, 2019; Le Meur et al., 2006; Milne, 2021; Nadeem et al., 2019).

Agriculture accounts for most (69%) of the global water withdrawals and 90% of consumptive water use (FAO Report, 2021; Shiklomanov, 2000; Siebert et al., 2010; UNESCO, 2020). Therefore, mapping irrigated fields globally is important for sustainable water use in agriculture (FAO Report, 2020). FAO's AQUASTAT dataset (<http://www.fao.org/aquastat>) provides global irrigation areas at a resolution of 5 arc-minutes. Thenkabail et al. (2009) produced a global irrigated area map at a nominal spatial resolution of 10 km using multiple satellite sensors. Based on the percentage of global irrigated area serviced by groundwater both in absolute terms and as a percentage of total irrigation, groundwater use is estimated to be increasing (Burke, 2002; FAO Report, 2021; Shah et al., 2007; Shiklomanov, 2000; Siebert et al., 2010; Thenkabail et al., 2009). However, these existing global irrigation area datasets are all at coarse spatial resolution. To the best of our knowledge, a global high-resolution and high-accuracy map for irrigation fields is still unavailable, although such an inventory is essential for agriculture management and planning. Gong et al. (2019) published a global high spatial resolution land cover dataset. However, there is only one main crop-related class in this dataset and no specific irrigated subclass data. Furthermore, the classification accuracy of "crop-land" in this dataset was not high (user's accuracy = 58.14%, and producer's accuracy = 61.07%). In fact, mapping global irrigated fields with a high accuracy using satellite imagery remains difficult under existing technical conditions because of the frequent confusion between irrigated fields and other ground land covers in remote sensing images (Bazzi et al., 2021; Gong et al., 2019; Thenkabail et al., 2009).

Center pivot irrigation systems (CPIS) are typical, popular modern irrigation systems that have been widely used around the world. Compared with traditional irrigation approaches, such as furrow irrigation and flood irrigation, CPIS has some notable advantages, such as more efficient water use, lower labor costs, and less soil tillage (New and

Fipps, 1976). This equipment is highly effective, especially over large areas.

Due to increasing food supply and food security stress, many countries have strong desires to increase their food production (Ahmed, 2020; Ingutia and Sumelius, 2022; Liu et al., 2021). Although there is plentiful evidence that using CPIS will increase the demand for freshwater, causing groundwater levels to decline and associated ecological problems (Chen et al., 2019c; Youssef et al., 2020), more and more farmers around the world are adopting advanced irrigation systems (e.g. CPIS) to increase crop productivity (Gathala et al., 2020; Levidow et al., 2014; Reportlinker, 2021). Farmers are encouraged because it is a direct and easy way to increase crop yield, especially in the context of capital influx from local incentive policies (German and Parker, 2018). In Zambia, for example, the number of CPIS has continued to grow over the past 15 years (German et al., 2020). The government's desire to increase crop productivity and improve the economy has brought about incentive policies for agriculture (e.g., low-interest rate loans for farmers, tax-cut policy for investments in agriculture), which have led to an influx of capital into agriculture (e.g., agricultural infrastructures, and advanced irrigation equipment) (Li and Wang, 2016; UNCTAD, 2011). These, in turn, have led to the adoption of CPIS in farmlands that previously used traditional furrow or flood irrigation, making crop production more water-efficient and more productive. In addition, woodland and shrubland have been converted to new CPIS-irrigated farmland (German et al., 2020).

It is estimated that the global CPIS market will reach \$3.9 billion by 2027, growing at a compound annual growth rate of 13.9% over the period from 2020 to 2027. This trend will certainly be helpful in increasing crop production in many countries. However, increasing CPIS could further intensify the water limitations in many arid areas around the world.

CPIS moves in a circular pattern, irrigating a circular area centered over the pivot. Therefore, when viewed from above, it generally creates a characteristic circular pattern in crops (those partly irrigated CPIS generally demonstrate circular segment shapes). Some example CPIS images are shown in Fig. S1. This makes them distinctive from other ground objects in remote sensing images. In the literature, some researchers have identified CPIS from remote sensing images by visual interpretation or using some automatic methods. For example, Rundquist et al. (1989) and Seth (2015) identified CPIS from Landsat imagery by visual interpretation. Tang et al. (2021) used circle Hough Transform algorithm to detect CPIS. In recent years, deep-learning techniques and algorithms, such as convolutional neural network (CNN), have grown in popularity and have been gradually adopted in remote sensing because they tend to have high accuracy for many applications (Ma et al., 2019). Zhang et al. (2018) first used CNN-based image classification method to identify CPIS from Landsat-5 TM images. Saraiva et al. (2020) used a semantic segmentation method to delineate CPIS from PlanetScope satellite images. Avery (2020), de Albuquerque et al. (2021a), and de Carvalho et al. (2021) used an instance segmentation method to detect and delineate CPIS from remote sensing imageries. However, these methods have only been tested in a few local areas. A global CPIS dataset is still unavailable, leading to a lack of systematic analysis of CPIS's impact on global water resources. As mentioned above, although a global map of all types of irrigated fields cannot yet be obtained accurately, CPIS, which is an indicator of industrial agriculture and large-scale investment in agriculture, can be reliably identified from remote sensing images and mapped because of its distinctive shape compared to other types of irrigated fields.

In this study, we used a CNN to map CPIS from Sentinel-2 satellite images, producing the first inventory of CPIS in global arid areas at a spatial resolution of 10 m. Based on these results, we evaluated the spatial relationship between CPIS, global surface water bodies and groundwater resources. We also mapped CPIS-equipped areas in 2000 using Landsat-7 satellite data, comparing the area to Sentinel-2 maps to estimate CPIS expansion over the first two decades of the new

millennium. As greater numbers of CPIS are expected to be installed worldwide and more and more traditional irrigated fields are equipped with CPIS, this global CPIS inventory will provide a better understanding of the current status of water use. This inventory can help governments in arid areas to judge better the environmental consequences of agricultural modernization and further help them make balanced agriculture and water management policies.

2. Materials and methods

2.1. Materials

Freely available Sentinel-2 and Landsat-7 data under an open data policy were used in our study. Sentinel-2 is an Earth observation mission in the Copernicus program coordinated and managed by European Space Agency. Sentinel-2 has a global coverage of land surfaces from 56°S to 82.8°N. The optical multi-spectral instrument on-board Sentinel-2 measures the ground reflected radiance in 13 spectral bands from visible to short-wave infra-red (spatial resolution of 10 m, 20 m and 60 m). Our study used the four bands with the highest spatial resolution (10 m) among the 13 available bands. Specifically, the blue (492 nm), green (560 nm), red (665 nm) and near infra-red (NIR) (833 nm) bands were used. From the Landsat-7 Enhanced Thematic Mapper Plus (ETM+), managed and operated by the United States Geological Survey, we used the first four bands in the range from visible to NIR, which have a spatial resolution of 30 m. Specifically, the blue (485 nm), green (570 nm), red (660 nm), NIR (840 nm) bands were used. For Landsat-7 analysis, pan-sharpened results were adopted using the panchromatic band (band 8) with a spatial resolution of 15 m.

The satellite data were mostly collected during local crop growing seasons to have a notable distinction between CPIS with its background land covers. In order to unveil the current distribution status of CPIS, we used the newest Sentinel-2 image among all the available multi-temporal data for each area during this study. A total of 11,378 Sentinel-2 scenes were used in our study. Most of the images used for the Southern Hemisphere were recorded in the second half of 2020 and the first half of 2021 (from September to February), and most of the images used for the Northern Hemisphere were recorded in 2021 (from April to September). The Landsat-7 data used were collected mostly in 2000, with a few in 1999 due to cloud cover. A total of 2739 Landsat-7 scenes were used in our study. The numbers of Sentinel-2 and Landsat-7 scenes used for each continent are shown in Table 1.

The global hyper-arid, arid and semi-arid map was derived from Global Aridity Index and Potential Evapotranspiration Climate Database v2 (Trabucco and Zomer, 2019), which provides high-resolution (30 arc-seconds) global aridity index data derived from global climate data for the 1970–2000 period.

Both the global extent of rivers and streams (Allen and Pavelsky, 2018) and the HydroLAKES datasets (Messenger et al., 2016) were used jointly in our study. The former provides global locations of rivers and

streams and their widths at a spatial resolution of 30 m. The latter provides locations and shapes of 1.42 million individual lakes and reservoirs globally that have areas of at least 10 ha. The world groundwater resources map (WHYMAP dataset) of 1: 25,000,000 (Richits et al., 2011) was used in our study, which provides various information on global characteristic groundwater environments, e.g., different hydro-geological provinces, groundwater recharge rates, etc. The dataset of world aquifer systems (WHYMAP and Margat, 2008), which provides the 37 largest global aquifer systems, was also used in our study.

The global environmental flow map (de Graaf et al., 2019), which presents where and when an environmental limit is reached resulting from groundwater pumping, was used in our study for data analysis. The GRACE total water storage anomalies (TWSA) dataset (Li et al., 2021; Save, 2020; Save et al., 2016) was also used in our study for data analysis.

2.2. Methods

2.2.1. Machine-learning method

In our experiments, a CNN-based instance segmentation framework was used. This method enables one to detect objects in an image while precisely delineating each distinct object with a mask. An input image first undergoes a stack of convolutional layers that have progressively coarser resolutions to generate a feature map. In our implementation, we used ResNeXt-101 (He et al., 2016; Xie et al., 2017) as the backbone network with a feature pyramid network (FPN) structure of four scale levels (Lin et al., 2017). The channel and spatial attention modules (Woo et al., 2018) were sequentially adopted for adaptive feature refinement. Then, a region proposal network (RPN) uses this feature map to generate bounding boxes that potentially contain an object. In the object detection branch, these candidate bounding boxes are fed into two additional fully connected sibling layers. One layer estimates the probability that the proposed object is a CPIS, and the other produces the coordinates of the CPIS's bounding box. In the segmentation branch, which is parallel with the object detection branch, a pixel-to-pixel segmentation mask is generated for each predicted CPIS (He et al., 2017). The segmentation result demarcates the boundaries of the detected CPIS. A cascade CNN-based instance segmentation framework (Cai and Vasconcelos, 2021) was used. This cascade network architecture consists of a sequence of detectors. Each stage aims to be more selective against close false positives for training the next stage by adjusting the bounding boxes to increase the detector quality at each stage with a progressively improved hypothesis. A three-stage cascade structure was used. In addition, the PointRend technique (Kirillov et al., 2020) was adopted to improve the segmentation result. A structure diagram of the used network architecture is shown in Fig. 1.

We first compared the performance of three state-of-the-art instance segmentation methods, which were originally proposed for common daily-life objects: Mask R-CNN (He et al., 2017), Hybrid Task Cascade (Chen et al., 2019a), and Cascade R-CNN (Cai and Vasconcelos, 2021). Note that the Mask R-CNN algorithm was used by Avery (2020), de Albuquerque et al. (2021a), and de Carvalho et al. (2021) to detect CPIS from Landsat-5, Sentinel-1 and Landsat-8 images, respectively. Because these off-the-shelf algorithms were originally proposed for detecting common daily-life objects and can only handle images that have three bands, we modified all of them by increasing the number of channels in the filters of their first convolutional layer to four (Chen et al., 2021). Then we further sequentially applied the convolutional block attention module (CBAM) technique (Woo et al., 2018) and the PointRend improvement technique (Kirillov et al., 2020) on the algorithm that achieved the best performance among the three methods.

In our implementation, the MMDetection toolbox based on PyTorch was used (Chen et al., 2019b). For the experimental hyperparameters, we set an Adam optimizer with a learning rate of 0.001 divided by ten after 1000 iterations and a momentum of 0.9. We used 20,000 iterations by keeping track of the validation loss to converge. We used 256 regions

Table 1

Number of used Sentinel-2 and Landsat-7 scenes for each continent, and the arid areas that were derived from the global aridity index and potential evapotranspiration climate database v2 (Trabucco and Zomer, 2019).

Continent	Arid areas (Mha)	Number of used Sentinel-2 scenes	Number of used Landsat-7 scenes
Africa	2039.50	3466	637
Asia	2209.35	4138	665
Europe	167.56	443	228
North America	757.58	1562	762
Oceania	694.05	1034	229
South America	389.59	735	218
World	6257.63	11,378	2739

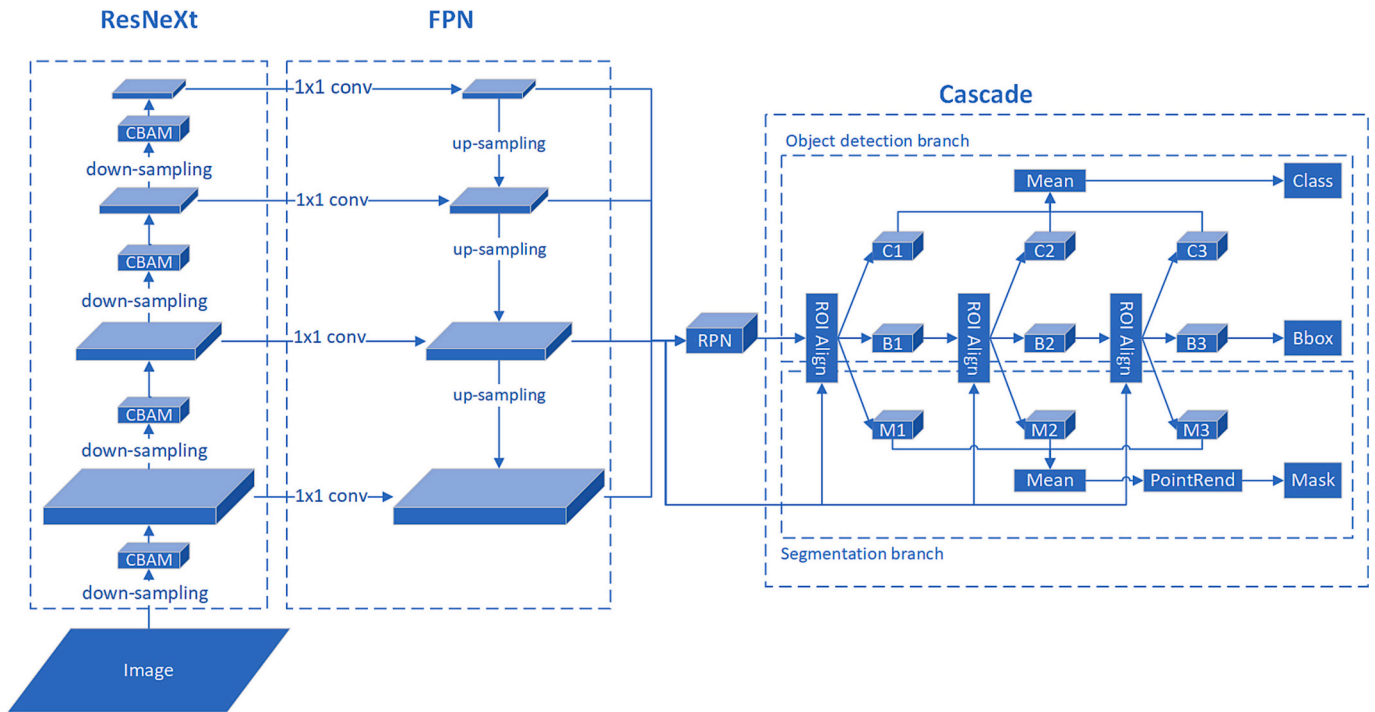


Fig. 1. Structure diagram of the used network architecture. An input image first undergoes a stack of convolutional layers to generate a feature map. Then, a region proposal network uses this feature map to produce proposals. In the object detection branch, these proposals are fed into two additional siblings, fully connected layers. One layer estimates the probability that the proposed object is a CPIS, and the other produces the coordinates of the CPIS's bounding box. In the segmentation branch, which is parallel with the object detection branch, a pixel-to-pixel segmentation mask is generated for each predicted CPIS. The cascade network architecture consists of a sequence of detectors. Each stage aims to be more selective against close false positives for training the next stage. Abbreviations: Conv, convolution operation; ResNeXt, residual network with next dimension; FPN, feature pyramid network; RPN, region proposal network; ROI, region of interest; Bbox, bounding box; Ci, classification probability at stage i ; Bi, bounding box at stage i ; Mi, mask at stage i ; CBAM, convolutional block attention module.

of interest for each image. Five anchor boxes of sizes 8, 16, 32, 64, 128 and three aspect ratios of 1:1, 1:2, and 2:1 were used by taking the sizes of CPIS into consideration. We used a cross-entropy function, an L1 loss function, and a cross-entropy loss function for the classification, bounding box regression, and mask segmentation tasks, respectively.

In our workflow, each Sentinel-2 scene was first divided into many 1000×600 sub-images as inputs for the CNN by considering available GPU memory size. These sub-images partly overlap with 100 neighboring pixels, but we detected the CPIS separately in each sub-image and merged the segmentation results. If a CPIS was located in an overlap region and was detected in multiple neighborhood sub-images, those multiply-detected results of the same CPIS were dissolved in the final merged results for the whole scene. The workflow for Landsat-7 is similar. A flowchart of the performed procedures is shown in Fig. 2.

We created a dataset of samples from five Sentinel-2 satellite images that include CPIS. The reference data for CPIS-covered area were carefully labelled with an open annotation tool of LabelMe software (<http://labelme.csail.mit.edu>). Note that we labelled a full circular shape for a CPIS even if it was partly vegetated when a clear full circular boundary of it could be observed. A total of 592 sample images were obtained, and each of them has a size of 1000×600 . Data augmentation

techniques such as flipping and zooming were used to enlarge the number of samples for training. We fine-tuned the used CNN model, which was pre-trained with the ImageNet dataset (<https://image-net.org>), with our CPIS training samples. Note that, as generally more samples used for training can ensure better machine-learning accuracy, all of the sample images from the five Sentinel-2 scenes, which were used to collect the dataset of CPIS samples, were used to re-train the CNN model for the other scenes worldwide.

2.2.2. Post-processing and visual check

We first converted the detection results from each Sentinel-2 scene into a shp file, and then dissolved them in ArcGIS®. The workflow for Landsat-7 scenes is similar. Although the automatically detected results have a precision higher than 90% (see Table 3 in the following), some false detections were also observed, which had similar shapes as CPIS, e. g., ponds that have a near-circular shape, and large bend vegetation fields. As most of these false detections are sparsely distributed, have irregular circular shapes, and are generally of smaller sizes than a typical circular of 400 m radius, we filtered them out by combining their roundness with their sizes. If an identified object either has a circularity value (defined as: $\text{circularity} = 4\pi \times \text{Area} / \text{Perimeter}^2$) less than a

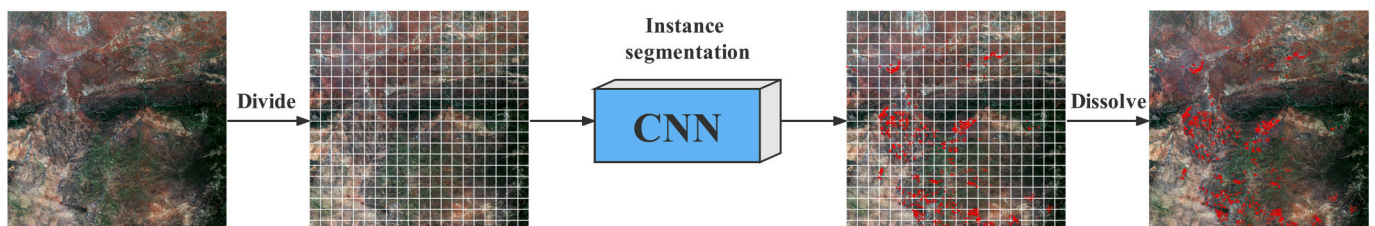


Fig. 2. Flowchart of the workflow. Note that the sizes of both sub-images and detected CPIS are enlarged to make them visible.

specified threshold or has an area less than a specified threshold, it would be sieved out. These candidate results were carefully visually checked by overlaying them on the satellite images, and those false detections were removed manually in the final inventory.

An erase geoprocessing operation was used to obtain the difference between the CPIS detection results in 2000 and 2021 in ArcGIS®. This difference result was overlaid on the Landsat-7 scenes to be visually checked the prior land use of them in 2000, which had been converted into agriculture fields equipped with CPIS in 2021.

2.2.3. Spatial analysis and used coordinate systems

We extracted the CPIS in each aimed study range (e.g., in each country, each continent, each aridity type, and each aquifer) using corresponding available shp files by clipping them from the whole obtained inventory. The numerical results (e.g., areas of CPIS, spatial distances from freshwater resources) shown in the paper and the Supplementary material were computed with Universal Transverse Mercator coordinate system in each zone and aggregated. Note that, for a clear demonstration of the global distribution of CPIS, most of the figures are presented in Plate Carree or Robinson projection systems.

We used a nearest-neighbor analysis for the CPIS with the surface water bodies. For each detected CPIS, the distance to its closest surface water body was calculated.

2.2.4. Model accuracy analysis

A four-fold cross-validation method was used to evaluate the performance of the trained CNN models (we combined the samples in two scenes that had fewer samples than the other three scenes). In each round, the samples in one scene were retained as the validation dataset to test the model, and the samples in the other scenes were used as training data. The average performance of the four rounds is reported. If a detected object has a score higher than a specified threshold, it will be regarded as a CPIS. The commonly used COCO metric (Lin et al., 2014) average precision (AP) was adopted to quantitatively evaluate the overall performance. Because AP is the mean performance derived with a precision-recall curve, in addition, the precisions AP75 and AP50 that correspond to threshold values of 0.75 and 0.5, respectively, were also used to evaluate the performance. The average recall (AR) was also used to evaluate the trained CNN models. These metrics are the most commonly used ones in instance segmentation tasks (Cai and Vasconcelos, 2021; de Albuquerque et al., 2021a; de Carvalho et al., 2021; He et al., 2017). The higher they are, the better the performance is.

2.2.5. Validation method of the obtained inventory

The validation design of the obtained CPIS inventory was performed with a random sampling technique of stratified ratio estimation with a set of sampling grid cells. The global arid areas were first divided into a grid of 10 km * 10 km sampling units. The grid cells that cover CPIS in the arid areas constitute our sampling frame. Globally, there are 24,879 grid cells that cover CPIS in arid areas. A two-layer strategy was used. The sampling frame was first divided into six strata by geographical continents. Then, each geographical stratum was further divided into three sub-strata by the covered area of CPIS in grid cells. Specifically, low-density ($C \leq 500$ ha), medium-density ($500 \text{ ha} < C \leq 1500$ ha), and high-density ($C > 1500$ ha), where C denotes the area that CPIS cover in each grid cell. Therefore, this two-layer stratified procedure resulted in 18 strata in our sampling frame. A total of 1290 stratified sampling test sites were randomly chosen worldwide to quantitatively validate the accuracy of the obtained dataset according to random sampling theory (Lohr, 2010). The sample size for each stratum was approximately allocated proportionally according to the number of sampling units in each stratum. At least 20 samples were used for a stratum (except for the small high-density strata in Oceania in which there are only eight samples that were used, and the small high-density strata in Europe in which 12 samples were used), and more samples for high-density strata than low-density strata. The numbers and areas of CPIS in samples were

counted, and the error was estimated using the ratio estimation method. The number of random validation samples for each stratum is presented in Table 2. The locations of validation sampling sites are shown in Fig. 3.

2.2.6. Comparison with other CPIS dataset

We compared the obtained CPIS inventory with the local Nebraska CPIS dataset in 2001 (<https://calmit.unl.edu/center-pivots-metadata-2001>). To the best of our knowledge, it is the only open large-scale CPIS dataset in arid regions, although it only covers parts of Nebraska, US. Although it is not at the same time point as our results in 2000, it still provides a valuable reference to validate our results because of the one-year time interval. We used a spatial analysis method for this comparison work. We overlaid our results in 2000 on the Nebraska dataset. If a CPIS in the Nebraska dataset intersected with a detected CPIS in our results in 2000, it was treated as being automatically detected successfully with our method.

3. Results

3.1. Model accuracy assessment

We compared the performance of three state-of-the-art instance segmentation methods, which were originally proposed for common daily-life objects. Three commonly used backbone networks, namely, ResNet-50 (He et al., 2016), ResNet-101 (He et al., 2016), and ResNeXt-101 (Xie et al., 2017), were compared. The results (Table 3, Fig. 4) indicate that the Cascade R-CNN algorithm with the ResNeXt-101 backbone network performed best among the three compared methods with various backbone networks. We then further sequentially applied the CBAM technique and the PointRend improvement technique on it in our ensuing experiments (denoted as proposed method in the following). It can be seen that both better precision and better recall were achieved when compared with the original Cascade R-CNN algorithm. Fig. 5 shows an example image of the instance segmentation performance of different methods. It can be seen that the proposed approach achieved the best performance. From Fig. 5(e), it can be seen that a more accurate segmentation result was achieved using PointRend technique.

3.2. Global CPIS distribution in arid areas

Overall there were about 10.26 million hectares (Mha) of irrigated areas equipped with CPIS in global arid areas in 2021, occupying 0.16% of the global arid areas. The current spatial distribution of CPIS in global arid areas is shown in Fig. 6. Although most CPIS are located in large plain lands, many separate small groups of CPIS occur in valleys. Despite the typical CPIS size of 50 ha with a circular radius of about 400 m, the largest CPIS covered 274 ha (with a circular radius of about 930 m) and the smallest CPIS covered only an area of about 6480 m² (with a circular radius of about 45 m). The densest CPIS-equipped fields were in the western US, South Africa, and Saudi Arabia.

By referring to the aridity climate zones derived from the global aridity index (Trabucco and Zomer, 2019), there were about 3.20 Mha (31.16%) of CPIS-equipped irrigation areas in arid regions and 6.30 Mha

Table 2

Number of validation samplings for each stratum. Low-density ($C \leq 500$ ha), medium-density ($500 \text{ ha} < C \leq 1500$ ha), and high-density ($C > 1500$ ha), where C denotes the area that CPIS cover in each sampling unit of 10 km * 10 km.

Continent	low-density	medium-density	high-density
Africa	202	40	20
Asia	154	40	20
Europe	59	20	12
North America	358	103	85
Oceania	70	20	8
South America	39	20	20
World	882	243	165

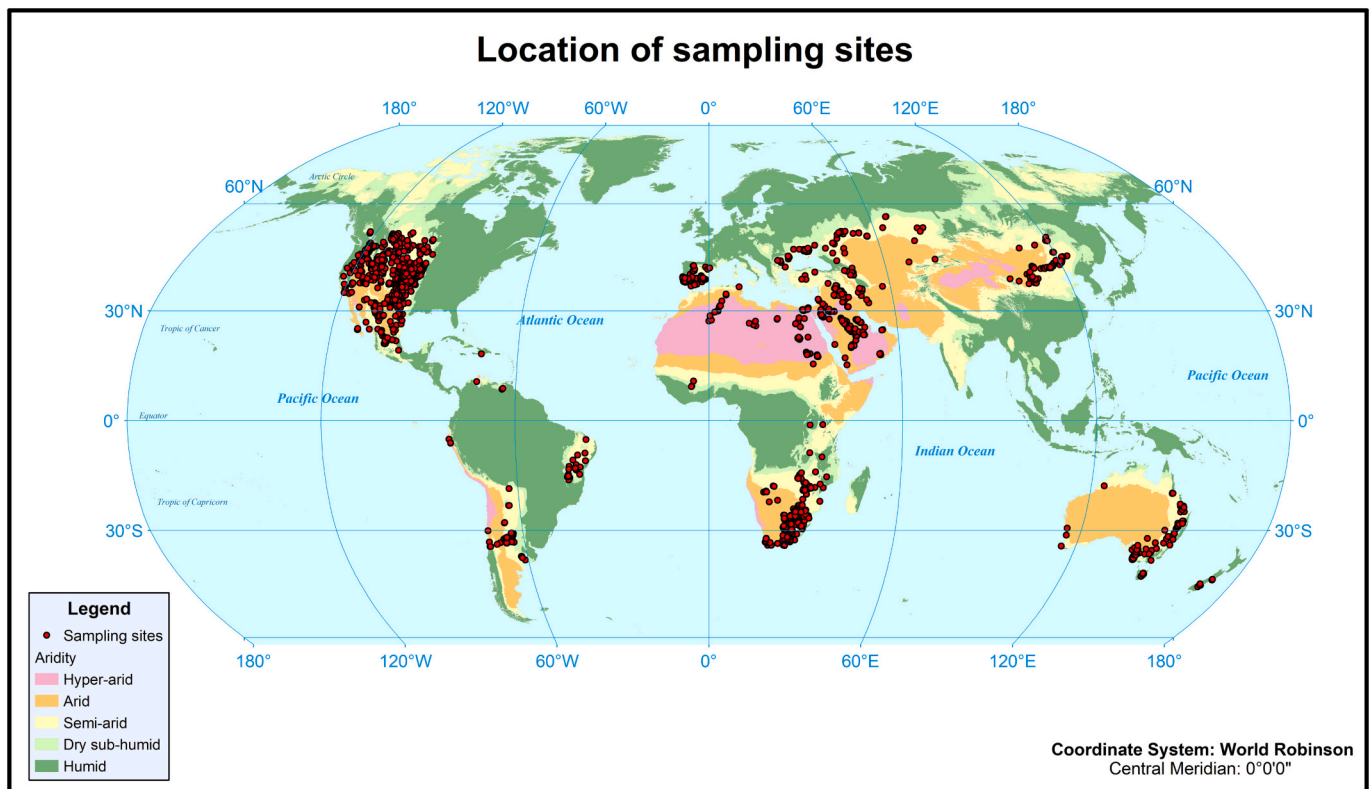


Fig. 3. Location of validation sampling sites.

Table 3

Model accuracy assessment (AP: average precision; AP₅₀, AP₇₅: precisions that correspond to threshold values of 0.5 and 0.75, respectively; AR: average recall).

Algorithm	Backbone	AP	AP ₅₀	AP ₇₅	AR
Mask R-CNN	ResNet-50	82.43	92.07	89.52	85.94
Mask R-CNN	ResNet-101	82.49	92.06	89.48	86.03
Mask R-CNN	ResNeXt-101	82.65	91.56	89.25	86.39
HTC	ResNet-50	83.11	91.96	89.20	87.29
HTC	ResNet-101	83.32	91.67	89.24	87.30
HTC	ResNeXt-101	83.93	92.51	89.77	87.53
Cascade R-CNN	ResNet-50	83.22	91.30	88.99	86.59
Cascade R-CNN	ResNet-101	83.25	91.60	89.35	86.67
Cascade R-CNN	ResNeXt-101	84.47	92.69	90.27	87.61
Cascade R-CNN + CBAM	ResNeXt-101	84.76	92.81	90.92	87.75
Cascade R-CNN + CBAM + PointRend	ResNeXt-101	85.63	93.76	91.53	89.04

(61.41%) in semi-arid regions (Table 4, Supplementary Fig. S2(a)). It should be noted that there are about 0.76 Mha (7.43%) of CPIS-equipped irrigation fields in hyper-arid areas (mostly in northern Africa and the Arabian Peninsula).

At the continental scale (Table 5, Supplementary Fig. S3(a)), most of the CPIS (6.96 Mha, 67.87%) were located in North America, occupying 0.92% of the arid areas of that continent. 1.41 Mha irrigation fields were equipped with CPIS in Asia (13.72%), 1.09 Mha in Africa (10.64%), 0.42 Mha in Europe (4.10%), 0.19 Mha in South America (1.88%), and 0.18 Mha in Oceania (1.78%).

At the country scale (Table 6, Fig. 7, Supplementary Table S1, Fig. S4 (a)), 75 countries in arid areas worldwide are using CPIS for irrigation. The largest acreage of CPIS-equipped irrigation fields is in US (6.268 Mha, 61.10%), followed by Saudi Arabia (0.908 Mha, 8.85%), South

Africa (0.563 Mha, 5.49%), Canada (0.359 Mha, 3.50%), Mexico (0.335 Mha, 3.27%), Egypt (0.207 Mha, 2.02%), China (0.192 Mha, 1.88%), Australia (0.180 Mha, 1.76%), Ukraine (0.177 Mha, 1.72%), and Argentina (0.135 Mha, 1.32%). More than 90% of CPIS-equipped irrigation fields in global arid areas are in these top ten countries.

Compared to CPIS-equipped irrigated areas mapped with Landsat-7 ETM+ data from 2000, we found a major expansion of CPIS worldwide. Overall there were 4.41 Mha of CPIS-equipped irrigated fields added in the past twenty years at a relative increase of 75.46% compared with those at the beginning of this century. This suggests increasing stress on water resources from CPIS-equipped irrigated fields in arid areas worldwide. By referring to the global aridity climate zones (Table 7, Supplementary Fig. S2(b), S2(c)), the largest increases in CPIS-equipped fields were in semi-arid regions (2.35 Mha). However, the relative increases in both hyper-arid (199.54%) and arid (94.43%) regions were actually larger than in semi-arid regions (59.56%). At the continental scale (Table 8, Supplementary Fig. S3(b), S3(c)), the largest extent of increased CPIS-equipped irrigated fields were in North America (2.37 Mha). However, the other five continents had higher relative increases, with Africa having the highest relative growth of 253.13%. On a country scale (Supplementary Table S1, Fig. S4(b), S4(c), Fig. S5), the US, Saudi Arabia, South Africa, Mexico, Egypt and China had the largest increase in acreage of CPIS-equipped fields. We also found 19 countries where CPIS were newly installed in arid areas over the past 20 years. CPIS-equipped fields in arid areas in 60 countries worldwide have more than doubled, indicating a current preference for using CPIS worldwide. We visually checked the detected CPIS results in 2000 and 2021 by overlaying them in ArcGIS®. The increased CPIS-equipped agriculture fields were obtained by differentiating these two results. By overlaying these difference results on the Landsat-7 scenes, we observed that much of the increased agriculture fields equipped with CPIS were placed in previously uncultivated lands (e.g., in Egypt, Algeria, Iraq, Saudi Arabia, South Africa), while the others were in fields where traditional irrigation was replaced by more advanced sprinkler irrigation.

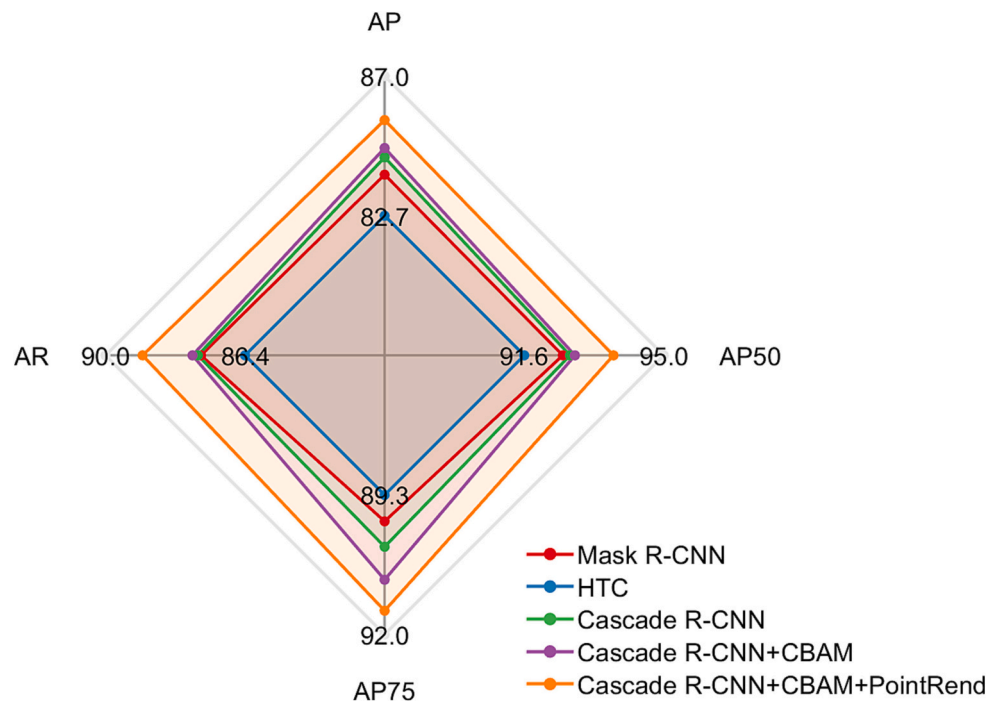


Fig. 4. Performance of different instance segmentation methods (AP: average precision; AP50, AP75: precisions that correspond to threshold values of 0.5 and 0.75, respectively; AR: average recall). Note that ResNeXt-101 was used for all of them.

3.3. Validation of the obtained CPIS inventory

The validation design of the obtained CPIS inventory was performed with the stratified ratio estimation sampling technique described in Sec. 2.2.5. With these sampling results, we estimated that the current true CPIS-equipped irrigated area in global arid areas is about 10.95 Mha, in the range of [10.88, 11.02] Mha with a 95% confidence level.

3.4. Comparison with other CPIS dataset

It was observed that 84% of the CPIS in the Nebraska dataset were automatically detected successfully. As the Nebraska CPIS dataset was obtained by manually digitizing Landsat-7 satellite images, our proposed automatic method reduces labor requirements. It achieves consistent and dependable CPIS results taking into account both the different time points and the fast annual increase rate of CPIS into consideration.

3.5. Spatial analysis with water resources

By referring to the world surface water maps (Allen and Pavelsky, 2018; Messenger et al., 2016), we found that most of the arid area CPIS are located far from surface water resources (Supplementary Fig. S6). Fig. 8 shows the cumulative percentage of CPIS to their nearest surface water bodies. Only 24.41% (2.50 Mha) of CPIS-equipped fields in arid areas were within 5 km of surface water sources, and about 57.95% (5.95 Mha) of all CPIS-equipped fields in arid areas had a distance >10 km from a water body with 1.29 Mha (12.52%) of the fields >50 km distant. The farthest CPIS was located 555 km from its nearest surface water body. Those CPIS that are far away from surface water sources are likely using groundwater for irrigation due to the high cost of surface water transport over long distances. Although infrastructure like aqueducts or canals are installed in some places, e.g., California, Arizona, Nebraska, US, for long distance surface water transport, there could also be heavy use of groundwater by CPIS that are far away from surface water sources. For example, by referring to the groundwater depletion map of the Ogallala aquifer (de Graaf et al., 2017, 2019), we found dense

CPIS to be highly correlated to groundwater depletion, independent of distance from surface water bodies (Supplementary Fig. S7). Note that for those CPIS that are close to surface water bodies, although surface water sources could be used, they could be augmented by groundwater, especially in some specific periods (e.g., dry years or during the dry season). Even if groundwater is not used, water pumping of these CPIS-equipped irrigated fields from nearby surface water resources still will have a major impact on surface water consumption, groundwater recharge, and downstream water supply and have detrimental effects on other competing water users.

Referring to the world groundwater map (Richts et al., 2011), we found that 1.43 Mha CPIS-equipped irrigated fields were located in arid areas that receive <200 mm/yr of rainfall and 1.59 Mha CPIS-equipped fields that have groundwater recharge below 100 mm/yr. Note that there are 0.07 Mha CPIS-equipped areas with groundwater recharge below 20 mm/yr, making these areas the most vulnerable to groundwater resource over-exploitation.

Fig. 9(a) shows the increased CPIS area in global arid regions from 2000 to 2021 at a spatial resolution of 0.5 arc-degree (a high-resolution image with zoomed-in sub-areas is shown in Supplementary Fig. S8). Referring to the GRACE total water storage anomalies (TWSA) data for this time period (Li et al., 2021; Save, 2020; Save et al., 2016), we found that most of the increased CPIS are installed in regions that have negative long-term TWSA (Fig. 9(b)). Large-scale densely installed CPIS from 2000 to 2021 may partly contribute to these negative TWSA and exacerbate the trend of declining groundwater resources in these areas (e.g., in the western United States, Arabian Peninsula, and northern Africa). By referring to the groundwater stress map derived from GRACE satellite in the world's 37 largest aquifer systems (Richey et al., 2015), we found that there is a total of 6.29 Mha CPIS-equipped irrigated fields, which are in the areas of 25 large aquifers (Supplementary Table S2). These CPIS-equipped irrigated fields could use groundwater extracted from these aquifers. Of these systems, the Ogallala aquifer has the largest coverage of CPIS-equipped irrigated fields (4.05 Mha). All five aquifers with the largest areas of CPIS-equipped irrigated fields, namely, the Ogallala, Arabian (0.91 Mha), Northern Great Plains (0.53 Mha), Karoo Basin (0.28 Mha), and Nubian (0.24 Mha), also show high or very

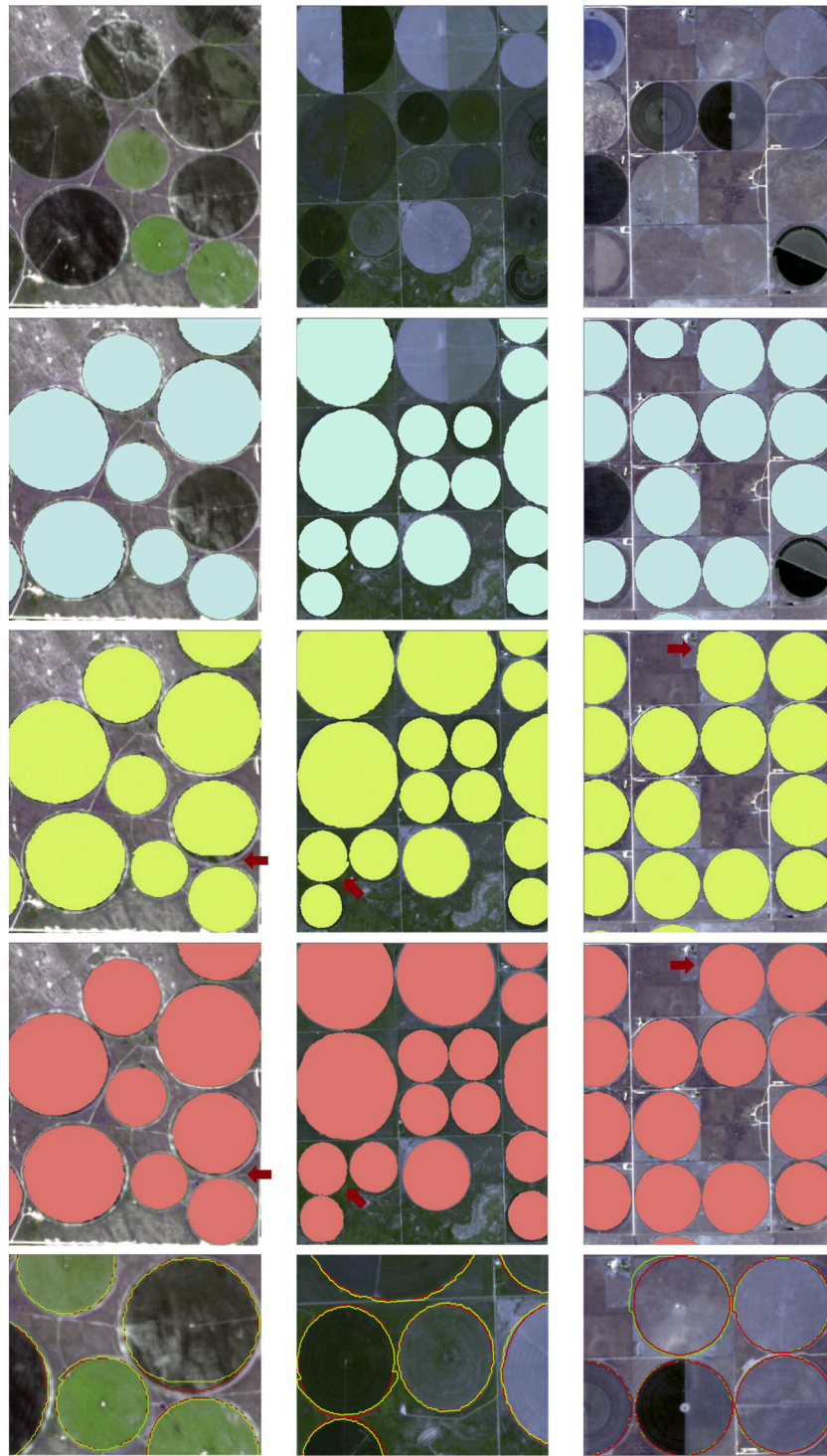


Fig. 5. Instance segmentation performance. (a) original image; (b) Cascade R-CNN; (c) Cascade R-CNN + CBAM; (d) Cascade R-CNN + CBAM + PointRend; (e) zoomed in the arrowed area in (c) and (d) by overlaying them together.

high groundwater stress (<https://www.un-igrac.org>). The high correlation between the groundwater withdrawal or the groundwater stress (derived from GRACE satellite or global groundwater model) and the area of CPIS-equipped irrigated fields indicates that the high rate of groundwater depletion is likely driven by anthropogenic irrigation demand from aquifers, including the Ogallala (de Graaf et al., 2017; Richey et al., 2015), Nubian (Frappart, 2020), Arabian (Sultan et al., 2019) aquifers.

By referring to the global environmental flow map (de Graaf et al.,

2019), there are 4.51 Mha (43.97%) CPIS-equipped irrigated fields in regions where environmental flow limits to groundwater pumping have already been reached, and there are 1.97 Mha (19.22%) CPIS-equipped irrigation fields in regions where environmental flow limits will be reached before 2040 (Fig. S9).

Numerous CPIS are installed near densely populated cities. Globally 339 cities are within 50 km of CPIS. Of them, 97 cities, e.g., Denver, Oklahoma City, Cairo, Johannesburg, Harare, Lusaka, Riyadh (Supplementary Fig. S10), have populations >500,000, and 152 cities have

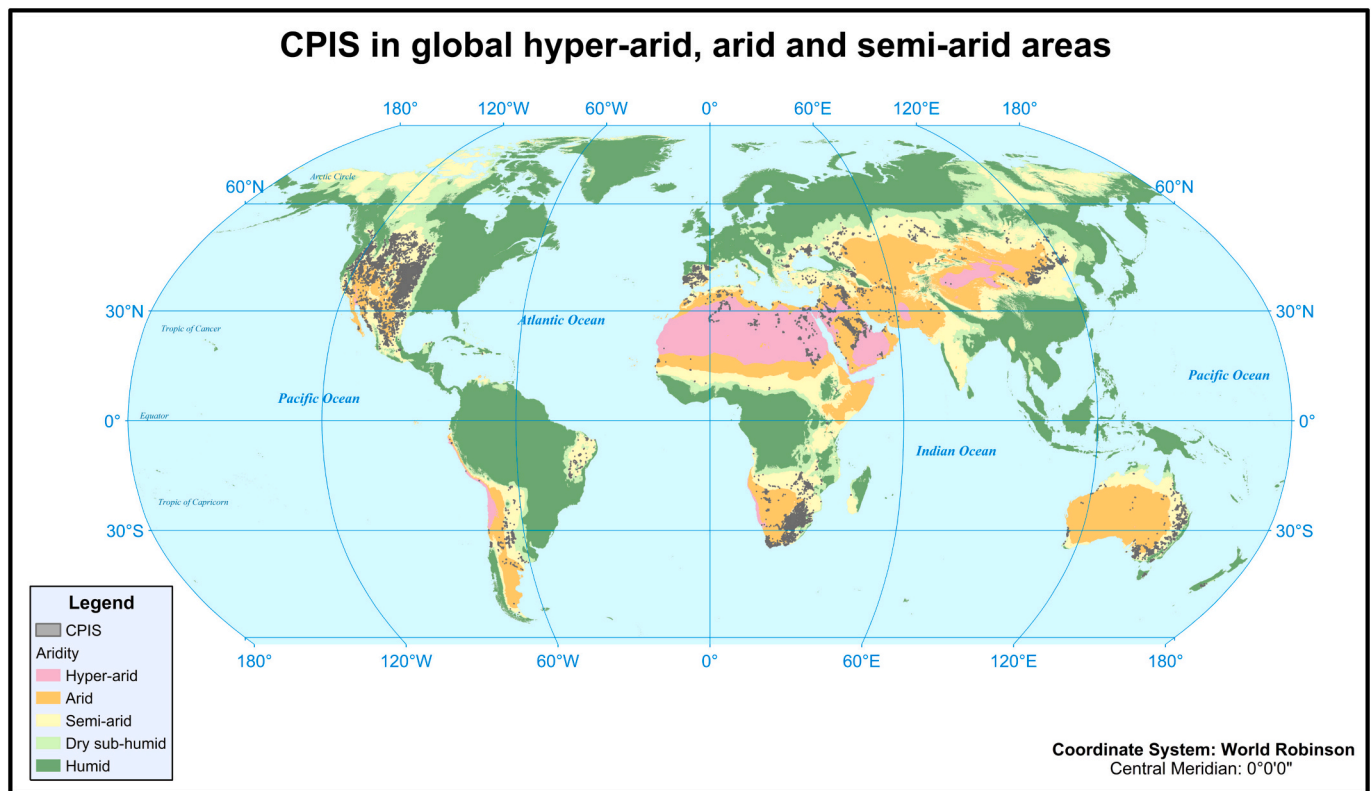


Fig. 6. Global distribution of CPIS in hyper-arid, arid and semi-arid areas.

Table 4

Global CPIS-equipped irrigation fields in hyper-arid, arid and semi-arid areas in 2021.

Aridity	Area (Mha)	Percent
Hyper-arid	0.76	7.43%
Arid	3.20	31.16%
Semi-arid	6.30	61.41%
Total	10.26	100%

Table 5

Global CPIS-equipped irrigation fields at continental scale in 2021.

Continent	Area (Mha)	Percent
Africa	1.09	10.64%
Asia	1.41	13.72%
Europe	0.42	4.10%
North America	6.96	67.87%
Oceania	0.18	1.78%
South America	0.19	1.88%

Table 6

Global CPIS-equipped irrigation fields at country scale in 2021.

Country	Area (Mha)	Percent
US	6.268	61.10%
Saudi Arabia	0.908	8.85%
South Africa	0.563	5.49%
Canada	0.359	3.50%
Mexico	0.335	3.27%
Egypt	0.207	2.02%
China	0.192	1.88%
Australia	0.180	1.76%
Ukraine	0.177	1.72%
Argentina	0.135	1.32%
Others	0.934	9.11%

populations above 100,000. Globally, Tabuk, Saudi Arabia, has the largest area (37,796 ha) of CPIS-equipped fields within 50 km to it (Supplementary Table S3), which is larger than the downtown area of this city. Many of these populated cities are partly dependent on groundwater and are already suffering urban water deficits (Flörke et al., 2018). Developing large-scale industrial agriculture near these cities will reduce the amount of priority water used for drinking purposes per capita due to increasing water competition between urban and agricultural demand. The increase of CPIS in these areas should be seriously evaluated, considering urban water security to supply drinking and hygiene water for daily lives.

There are 0.13 Mha CPIS-equipped irrigation fields in coastal saline groundwater areas (WHYMAP and Margat, 2008) (e.g., in north-eastern Egypt, south-eastern Saudi Arabia, southern Australia, and southern Argentina). This suggests that these areas are more susceptible to salt-water intrusion than other areas, and the situation could deteriorate with the development of irrigation intensification.

4. Discussion

4.1. Error sources of the obtained CPIS inventory

As almost all of the false detections in the obtained inventory were visually checked and manually removed, it was observed that the commission error in this inventory was almost fully removed, and only omission errors remained. The main reasons for this omission error could be: 1) Some CPIS have irregular shapes differing from a common circle. 2) Some CPIS were in abandoned or fallow states, which generates low contrast with their backgrounds. 3) Although we labelled a full circular shape for a CPIS even if it is partly vegetated in the training dataset and our method was observed to successfully obtain a full circular shape for most of them, there are still some circular CPIS that were not fully segmented (Supplementary Fig. S11). 4) Although the Sentinel-2 data with the least cloud cover was used (a <10% cloud cover

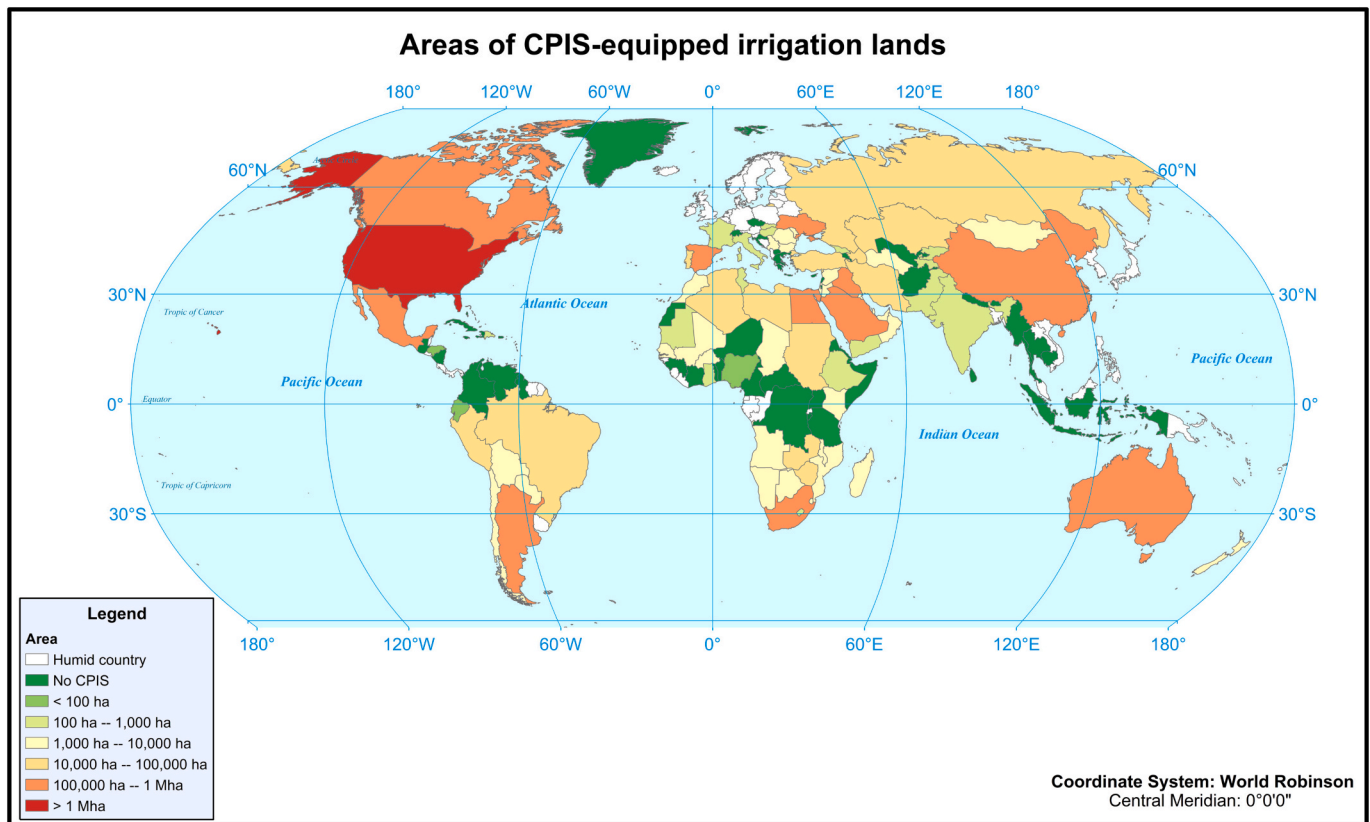


Fig. 7. Area of CPIS-equipped irrigation lands at the country scale in 2021.

Table 7

Relative increase of global CPIS-equipped irrigation fields in hyper-arid, arid and semi-arid areas from 2000 to 2021.

Aridity	Area of CPIS-equipped lands (Mha)		
	2021	2000	Relative increase (%)
Hyper-arid	0.76	0.26	199.54
Arid	3.20	1.64	94.43
Semi-arid	6.30	3.95	59.56
Total	10.26	5.85	75.46

Table 8

Relative increase of global CPIS-equipped irrigation fields at continental scale from 2000 to 2021.

Continent	Area of CPIS-equipped lands (Mha)		
	2021	2000	Relative increase (%)
Africa	1.09	0.31	253.13
Asia	1.41	0.56	149.71
Europe	0.42	0.24	78.98
North America	6.96	4.60	51.46
Oceania	0.18	0.06	183.90
South America	0.19	0.08	151.12

threshold was used), some separate clouds and their shadows were still cast on CPIS. Fig. S12 shows a few samples that were not successfully detected or segmented by our method. Multi-temporal images can be used to reduce some of these interferences, such as cloud cover and shadow (de Albuquerque et al., 2021b).

Although we selected local crop growing seasons preferably to ensure there is a relatively high amount of vegetation in the farmlands, it was observed that some detected fields did not have any vegetation but still demonstrated clear contrast with their backgrounds. Some CPIS-

covered fields were partially vegetated, and it is hard to ascertain whether the non-vegetated areas were being utilized or not. Relying solely on remote sensing data is insufficient to infer such information, and fieldwork could be necessary. It is an open problem of using a single image for crops that could have seasonal planting variation over a whole year. Using multi-temporal images in a year with time series of vegetation indices could provide a partial solution to this problem. This will be addressed in future work.

4.2. Various advanced irrigation technologies

Although intelligent irrigation equipment for more efficient water and energy savings is desired and being developed, CPIS has already proven to provide increased crop productivity and conserve more water than traditional flood or furrow irrigation systems (New and Fipps, 1976). It is unlikely that there will be a better alternative widely used in the near future. Drip irrigation is another water-efficient system, but it is costly and requires extensive maintenance (Lamm and Camp, 2007; Surendran and Chandran, 2022). Plastic mulching can reduce water loss from soils due to evaporation, but it is not suitable for all crop types and has environmental problems (Kyrikou and Briassoulis, 2007). In fact, it is likely that more and more CPIS will be installed worldwide (Reportlinker, 2021).

Although a trade-off between maximizing crop water productivity (CWP) with the expense of crop yield reduction is advocated, individual farmers in poorly regulated markets generally prefer to use more water to maximize crop yield than to sacrifice their economic benefits (Letseku and Grové, 2022; Sanderson and Hughes, 2019). Technological advances in water efficiency would not naturally guarantee a reduction in overall water consumption, as the benefits of water efficiency will soon be overwhelmed by the expanded irrigation that follows the lower cost and increased income (Pfeiffer and Lin, 2014; Sanderson and Hughes, 2019). With the increasing decline of groundwater levels, in regions

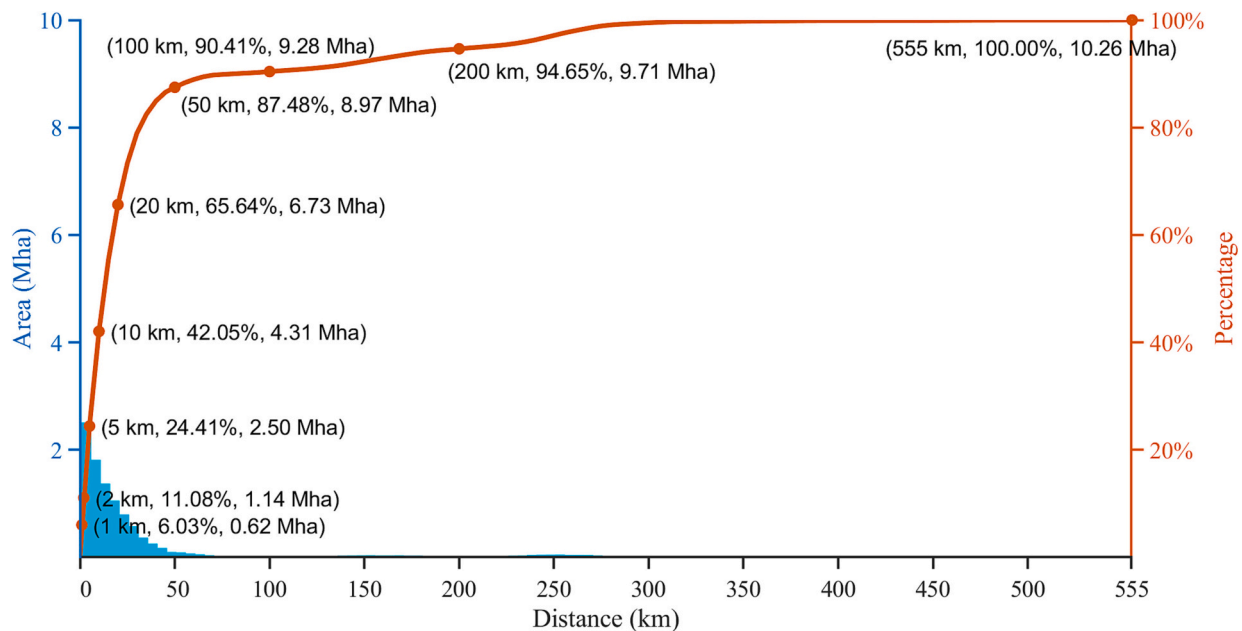


Fig. 8. Cumulative percentage of CPIS to their nearest surface water bodies.

with insufficient water and agricultural management, private schemes might prefer to drill deeper to extract more groundwater than their neighboring competitors rather than accepting a trade-off between maximizing CWP with crop yield reduction. Grain production depends on who has water and who does not. Excessive water extraction by competing water users will place freshwater resources under increasing stress and soon exceed water availability. Therefore, if there is no regulation on the total amount of water extraction, even adopting more advanced water-saving agricultural technologies would not necessarily reduce the total consumption of water for irrigation.

4.3. Intensifying competition from irrigation for water

Our results reveal the current status of irrigation in global arid areas and its development in the first twenty years of the 21st century, which indicates an intensifying competition from irrigation for water use in those areas. It is noted that except for the US, most of the newly increased CPIS worldwide were installed in low-income countries (IMF, 2022), which are facing increased food demand and increasing water shortages. The global expansion of CPIS-equipped irrigation areas indicates that the stress on water resources from irrigation is increasing worldwide. Referring to FAO statistical data of global irrigation areas (Siebert et al., 2013), North America and Africa had the two highest CPIS-equipped percentages in arid areas, covering about 19.12% and 8.06% of all irrigation areas, respectively.

Although at a global scale the water demand from irrigation could be different in various regions due to many natural and anthropogenic factors, the area of irrigation fields is an important factor involved in water demand from agriculture. Note that, developing a global hydrological model by considering all those factors is beyond the scope of this remote sensing paper. Nevertheless, the obtained inventory could be used as an input for some existing hydrological models. If other factors (e.g., crop type, soil type, water usage efficiency) are assumed to be relatively stable at the local scale, the increased CPIS area would be highly correlated with the water demand from irrigation locally. By taking both climate conditions and increased irrigation area into consideration, overall, the high increased stress on water resources from CPIS in southwestern North America could be mainly due to the irrigation equipment densely installed in those semi-arid areas. Similarly, northern Africa and the Arabian Peninsula could be highly stressed for

water sources due to both increased dense irrigation fields and the hyper-arid climate.

4.4. Balance between freshwater preservation and agriculture irrigation

Unsustainable groundwater extraction will lead to detrimental effects (UNESCO, 2020, 2021; Gleeson et al., 2020), and these issues will become more complex with global climate changes and expected changes in precipitation and evaporation (Wu et al., 2020). Although CPIS may not be the sole factor for the decline of the water table in the affected arid areas, intensifying agricultural water use in these areas, which is indicated by capital-intensive and high-input high-output CPIS, may still play an important role that cannot be ignored. As some freshwater resources are transboundary, especially for over-exploited freshwater aquifers with limited recharge, it highlights the need for concerted global transboundary water management action by those governments that share water resources. In fact, as the full effects of groundwater depletion are delayed compared to surface water, it may take decades or centuries to manifest the detrimental impacts of groundwater overdraft, though some regions around the world are already severely affected. Groundwater pumping for agriculture irrigation is necessary, but what really matters is the scale of pumping, especially in those arid areas where groundwater is dwindling at an alarming rate, and an unfolding crisis looms. In the long term, groundwater overdrafts will make irrigation less profitable and cause ripple effects throughout the economy. Some governments would be facing a water bankruptcy state, which will threaten their natural environment and future socio-economic development.

It is not reasonable to prohibit or decrease irrigation in some arid countries at the current stage, as the food demand with a growing population is urgent. Nevertheless, from the perspective of ecological sustainability of water resources, we suggest the need for a balance between water use and agriculture development. Such an equilibrium could be achieved by socio-economic organizational measures at the national or regional level, for example, encouraging the use of less water-intensive crops in arid areas; subsidizing more water-efficient but more costly technologies (e.g. drip irrigation); regulating the use of treated water for irrigation; charging for water-overuse to incentivize farmers to adopt more water-saving practices; water trading; restructuring agriculture across different regions that have different

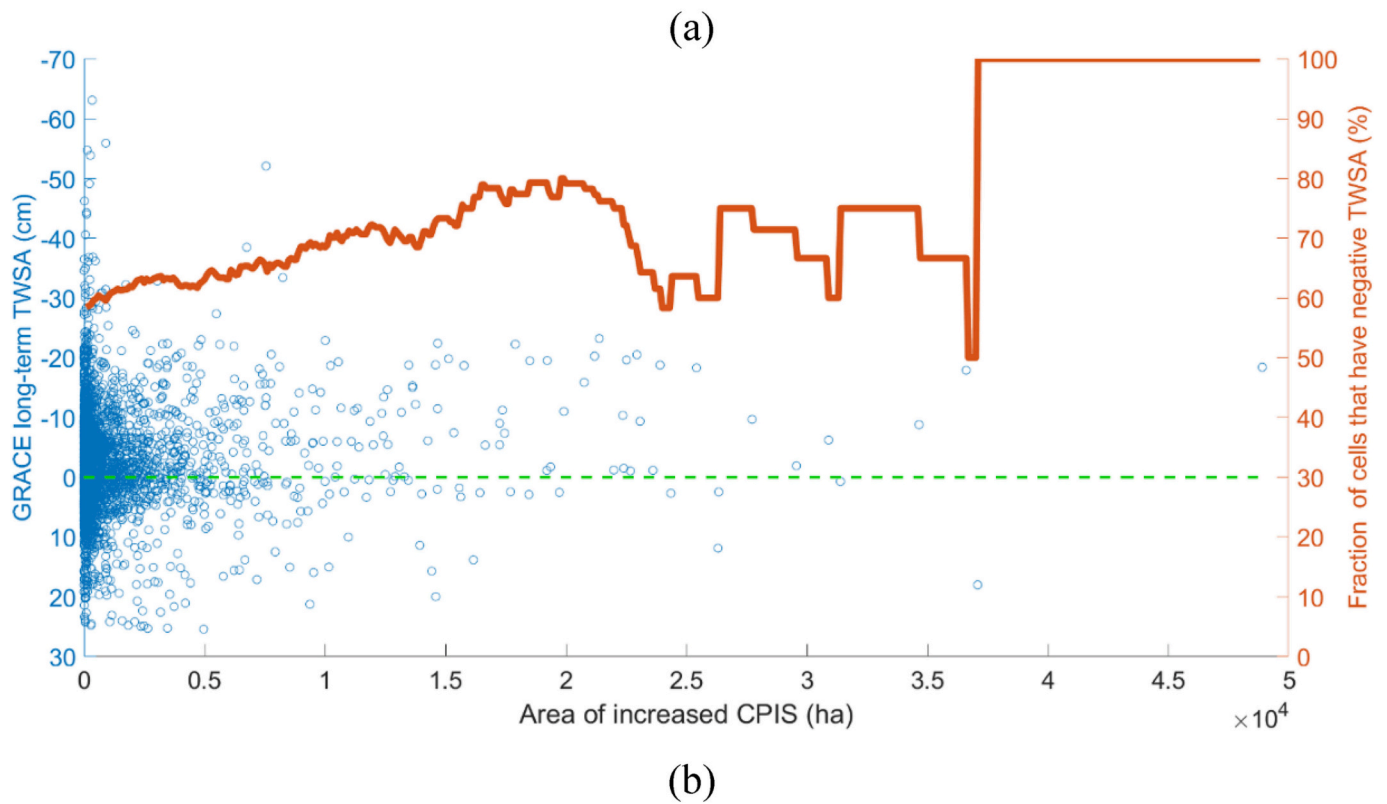
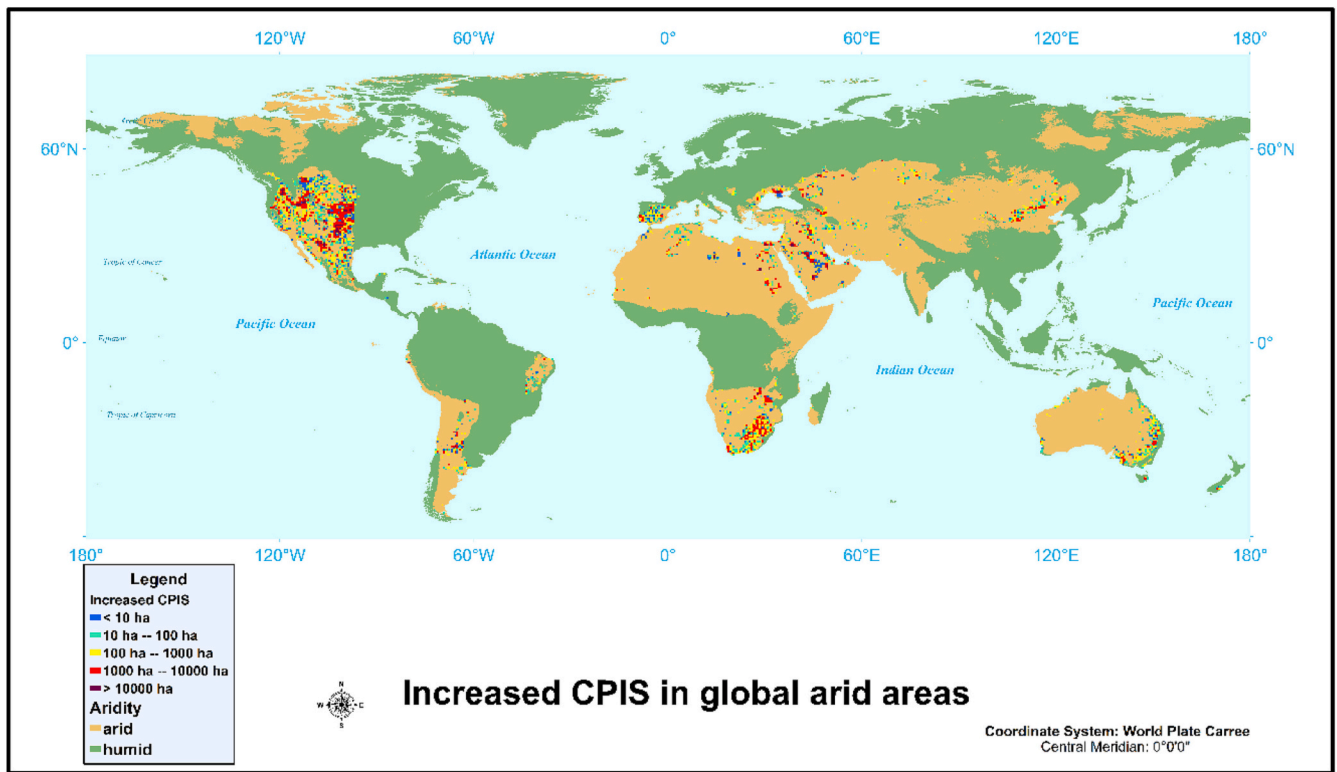


Fig. 9. Increased CPIS area and spatial analysis with GRACE data. (a) Increased CPIS area in global arid regions from 2000 to 2021 at a spatial resolution of 0.5 arc-degree; (b) The x-axis gives the area of increased CPIS. The left y-axis shows the GRACE long-term TWSA (blue points). Note that the left y-axis was flipped. The right y-axis shows the fraction of cells that have negative TWSA (brown solid line). A TWSA value of 0 is shown in dashed green line, below which the points show groundwater declines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

water conditions; re-allocating water rights for different parts, etc. These socio-economical organizational innovations might be more important than sole water-efficient technological ones.

Unchecked anthropogenic depletion of water sources will foreclose the options for future growth for many countries. Irrigation is not only a major cause of water scarcity but also will be a victim of water scarcity. Nonetheless, the unwillingness of farmers to compromise short-term gain for long-term sustainability should be considered when making related agriculture and water policies. More and more governments are facing the challenge of balancing increasing water demands for agriculture, urban, industrial, and environmental uses. Irrigation consumes most of the pumped water resources. Hence, industrial agriculture expansion should be within a framework of the water-energy-food security nexus to properly manage difficult trade-offs and promote sustainability. Economic development cannot be built and maintained without enough water. To rationally utilize natural freshwater resources and for sustainable development, there should be a well-considered balance between freshwater preservation and agriculture irrigation in arid areas developed as soon as possible.

4.5. Contributions and limitations

This paper makes three major contributions: 1) Remote sensing of agriculture is an efficient surveillance and mapping technique compared to in situ or on-site observation. We proposed to map CPIS, which is an important indicator of modern large-scale industrial agriculture, using a deep-learning technique and adopted two approaches to improve the performance of the off-the-shelf instance segmentation algorithm. The proposed method provides a fast, low-cost, and accurate solution for mapping CPIS fields, which can facilitate constant irrigation monitoring at both local and global scales. 2) Our results revealed the current status of irrigation in global arid areas and its development in the first two decades of the 21st century. Through spatial analysis of freshwater resources, our results also revealed that intensifying freshwater (especially groundwater) demand from irrigation is not necessarily an issue related to a specific country but a global problem. 3) The obtained CPIS inventory that we have made publicly available can serve both agricultural and hydrological researchers with precise irrigation data at a global scale. It also can be used by both local agriculture and local water management sectors. It can provide governments with accurate irrigation data, and help them understand and mitigate conflicts between freshwater preservation and agriculture irrigation, which in turn helps making better balanced agriculture and water management policies.

This paper has certain limitations, which are as follows: 1) We have only mapped the CPIS in global arid areas for two specific years, which are 2000 and 2021. There is no detailed annual CPIS information for each year in this period, and there is also no CPIS information beyond it. Note that, when considering historical data for mapping CPIS, the Landsat series satellites have a much more extensive archive of past recordings than the Sentinel-2, which was launched in June 2015. However, for some years after 2003, the problem of scan line failure in Landsat-7 ETM+ could be a significant issue that needs to be considered. 2) Although the spatial resolutions (10 m of the Sentinel-2 and 15 m of the Landsat-7 pan-sharpened data) suffice to map CPIS that has a typical circular radius of 400 m, improved accuracy of segmentation results may be possible by using other data with finer spatial resolution. 3) Although the performance of the proposed method may be satisfactory for many applications, considering that deep-learning technology is evolving rapidly, the accuracy could be further improved by testing other state-of-the-art instance segmentation methods. These issues will be addressed in our future work.

5. Conclusions

CPIS are indicators of intensified industrial agriculture and large-scale investment in agriculture. This study mapped CPIS with satellite

data using machine learning in global arid areas and analyzed the spatial relationships between CPIS and freshwater resources. The results indicate that most of the CPIS likely operate by withdrawing groundwater, and the stress on freshwater resources is intensifying in arid areas globally in the past twenty years. This paper provides insight from the view of water resources sustainability on the global development of agricultural intensification in arid areas and the stress from irrigation. The outcomes can help governments to better understand the current status of freshwater stress and ecosystem sustainability in relation to the development of agricultural intensification, and further to make balanced agriculture and water management policies.

Code availability

Some example codes are available in an online repository at https://github.com/DetectCPIS/global_cpis_codes, where future updates will be implemented. The full codes developed for this research are available on request to the corresponding author.

Credit author statement

Fen Chen: Conceptualization, Methodology, Software, Analysis, Validation, Writing-Original draft preparation, Writing-Reviewing and Editing

Haojie Zhao: Methodology, Software, Analysis, Data curation, Visualization

Dar Roberts: Methodology, Analysis, Writing-Reviewing and Editing

Tim Van de Voorde: Analysis, Writing-Reviewing and Editing

Okke Batelaan: Writing-Reviewing and Editing

Tao Fan: Software, Data curation

Wenbo Xu: Resources

Declaration of Competing Interest

The authors declare no competing financial interests.

Data availability

Our global CPIS inventory is publicly available in an online repository at https://github.com/DetectCPIS/global_cpis_shp. All satellite data used in this study are freely available under the open data policies adopted by ESA and by USGS.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rse.2023.113760>.

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